Skills Summary

Putting Patterns, Data and Points Together

Use this reference while completing your worksheet or when studying.

Skill 1 — Jumping to any term of a pattern

Rule: The first term is the start, so term n is $n-1$ steps out.

term n = start + step size $\times(n-1)$

Two patterns meet when their terms are equal; the one that starts behind closes the gap by the difference of the step sizes each step.

Example: A pattern starts at 5 and adds 7 each time. Write the 9th term.
Step 1. Listing nine terms invites a miscount, so I count steps instead: the 9th term is eight steps past the start.
Step 2. $5+7\times 8=5+56$. 9th term: **61**

Try it: A pattern starts at 2 and adds 9 each time. Write the 7th term.

check: **56**

Skill 2 — Counting from a line plot

Rule: On a line plot the tick is the *value* and the stack height is *how many*.

"How many students" counts crosses. "How many books" uses the numbers under them. Comparing two stacks means subtracting their heights, not their ticks.

Example: A plot has 3 crosses at 1, 5 crosses at 2, 2 crosses at 3 and 4 crosses at 4. How many entries sit at value 2?
Step 1. The question asks how many, not how much, so I read the height of the stack above the tick and ignore the tick itself.
Step 2. The stack above the tick labelled 2 is 5 crosses tall. Entries at value 2: **5**

Try it: On the same plot, how many more entries sit at value 2 than at value 3?

check: **3**

Skill 3 — Turning a pattern into ordered pairs

Rule: Pair the term number with the term value: (term number, value).

Across is the term number, up is the value. Because the step never changes, the marks always land on a straight line — so a bent line is a signal that a term was generated wrongly, not a new shape.

Example: A pattern starts at 0 and adds 3. Write the ordered pair for term 7.
Step 1. The term number is already the across coordinate, so the only work is generating the value that goes up.
Step 2. Term 7 is six steps out: $0 + 3 \times 6 = 18$. Ordered pair: $(7, \mathbf{18})$

Try it: A pattern starts at 1 and adds 5. Write the ordered pair for term 6.

check: $(6, \mathbf{26})$

Skill 4 — Totalling a whole plot

Rule: Multiply each value by its stack height, then add the products.

total = (height × value) + (height × value) + . . .

Once the total is known, mean = total ÷ number of entries — and running that backwards (mean × number of entries) is a genuine check on the total.

Example: A plot has 3 crosses at 1, 2 crosses at 4 and 5 crosses at 2. What is the total?
Step 1. Adding the ticks alone would ignore how many entries sit on each, so I weight every value by its stack height first.
Step 2. $3 \times 1 + 2 \times 4 + 5 \times 2 = 3 + 8 + 10$. Total: **21**

Try it: A plot has 2 crosses at 3, 4 crosses at 5 and 1 cross at 6. What is the total?

check: **32**